Shamba ShapeUp – Full Solution Documentation & Validation Report

---

1. Executive Summary

This solution predicts **future purchasing behavior** for each customer–product combination using historical weekly transaction data. Specifically, the model forecasts:

Classification Targets

- Purchase next 1 week → `Target_purchase_next_1w`
- Purchase next 2 weeks → `Target_purchase_next_2w`

Regression Targets

- Quantity next 1 week → `Target_qty_next_1w`
- Quantity next 2 weeks → `Target_qty_next_2w`

The solution combines:

- Advanced time-series feature engineering
- Strict leakage prevention using gap-aware validation
- Horizon stacking for multi-step forecasting
- Dual-model ensemble (LightGBM + CatBoost)

2. High-Level Architecture

The solution pipeline:

Raw Data
↓
Data Merging (SKU + Customer)
↓
Time-Aware Feature Engineering
↓
Horizon Stacking (H1 + H2 unified training)
↓
Gap-Aware Cross Validation
↓
Dual Model Training (LGB + CatBoost)
↓

Ensemble Predictions

3. Data Sources

Input Files

- Train.csv — Weekly historical observations
- Test.csv — Future prediction rows
- sku_data.csv — Product metadata
- customer_data.csv — Customer metadata

Merge Strategy

SKU merged using:

- product_unit_variant_id
- unit_name
- grade_name

Customer merged using:

- customer_id

4. Feature Engineering

All features are generated using **full chronological history** after concatenating train and test, ensuring correct time ordering.

4.1 Lag Features

Generated for:

- qty_this_week
- num_orders_week
- spend_this_week
- purchased_this_week
- Target_purchase_next_2w
- Target_qty_next_2w

Lag windows:

2–8 weeks

Purpose:

- Capture recent purchasing behavior

- Encode recency patterns

4.2 Recency & Last Purchase Features

Includes:

- Weeks since last purchase
- Last purchase quantity
- Last purchase spend

Purpose:

- Capture purchase cycles and repeat behavior patterns

4.3 Rolling Window Features (Leakage Safe)

Windows:

- 4 weeks
- 8 weeks

Statistics:

- Mean
- Std
- Min
- Max

Leakage Prevention:

- Most features use shift(1)
- 2-week targets use shift(2)

Purpose:

- Capture short-term trends
- Capture volatility

4.4 Expanding Features

Statistics:

- Expanding mean
- Expanding std
- Expanding min/max

Purpose:

- Capture long-term baseline behavior

4.5 Trend Features

Examples:

- Rolling mean now − rolling mean N weeks ago

Purpose:

- Capture demand momentum (increasing / decreasing)

4.6 Calendar Features

Extracted from week_start:

- Year
- Month
- ISO Week
- Quarter

Purpose:

- Capture seasonality patterns

4.7 Customer Lifecycle Features

- Customer tenure in days
- New customer flag (< 30 days)

Purpose:

- Capture onboarding and early lifecycle behavior

4.8 Global Aggregation Features

Computed at:

- Customer level
- Product level
- Product variant level

Aggregations:

- Mean
- Sum

Purpose:

- Provide strong baseline priors

4.9 Relative Ratio Features

Examples:

- Qty vs customer average
- Spend vs customer average
- Qty vs product average

Purpose:

- Normalize behavior relative to baseline

4.10 History Depth Features

- History length in weeks
- Low history flag (< 4 weeks)

Purpose:

- Improve cold-start handling

5. Leakage Prevention Strategy

Strict temporal integrity maintained via:

Shifted Feature Computation

Ensures only past data influences current predictions.

Gap-Aware Validation

Validation folds simulate real forecasting scenario:

Train History → Gap Week → Horizon 1 → Horizon 2

Feature Masking During Validation

- Horizon 1 → mask gap-week lag features
- Horizon 2 → mask gap-week + horizon-1 lag features

6. Modeling Strategy

Two models per task:

LightGBM

Strengths:

- Fast training
- Strong tabular performance

CatBoost (GPU)

Strengths:

- Native categorical handling
- Strong regularization
- Robust generalization

7. Horizon Stacking

Instead of training separate models per horizon:

Dataset duplicated with feature:

horizon $\in \{1, 2\}$

Benefits:

- More training data
- Shared pattern learning
- Better generalization

8. Ensembling Strategy

Final predictions:

Final = 0.5 × LightGBM + 0.5 × CatBoost

Benefits:

- Reduced variance
- More stable predictions

9. Validation Metrics Report

Validation Setup

3-fold time-aware validation with 1-week gap simulation.

Fold-Level Results

Fold 1

Horizon 1:

- LGB: LogLoss 0.0576 | AUC 0.9798 | MAE 3.0074
- CAT: LogLoss 0.0717 | AUC 0.9815 | MAE 3.1325
- ENS: LogLoss 0.0625 | AUC 0.9814 | MAE 3.0579

Horizon 2:

- LGB: LogLoss 0.0726 | AUC 0.9686 | MAE 3.3546
- CAT: LogLoss 0.0920 | AUC 0.9720 | MAE 3.6078
- ENS: LogLoss 0.0790 | AUC 0.9721 | MAE 3.4464

Fold 2

Horizon 1:

- LGB: LogLoss 0.0589 | AUC 0.9755 | MAE 0.7590
- CAT: LogLoss 0.0713 | AUC 0.9774 | MAE 0.8071
- ENS: LogLoss 0.0628 | AUC 0.9771 | MAE 0.7499

Horizon 2:

- LGB: LogLoss 0.0703 | AUC 0.9709 | MAE 3.3248
- CAT: LogLoss 0.0889 | AUC 0.9612 | MAE 3.5565
- ENS: LogLoss 0.0760 | AUC 0.9683 | MAE 3.3998

Fold 3

Horizon 1:

- LGB: LogLoss 0.0588 | AUC 0.9682 | MAE 2.6956
- CAT: LogLoss 0.0724 | AUC 0.9692 | MAE 2.8129
- ENS: LogLoss 0.0633 | AUC 0.9690 | MAE 2.7161

Horizon 2:

- LGB: LogLoss 0.0686 | AUC 0.9743 | MAE 3.7262
- CAT: LogLoss 0.0825 | AUC 0.9755 | MAE 3.9864
- ENS: LogLoss 0.0729 | AUC 0.9758 | MAE 3.7956

Final Training Iterations

LightGBM:

- Classifier → 230
- Regressor → 177

CatBoost:

- Classifier → 550
- Regressor → 1327

Key Observations

- Horizon 1 consistently performs better than Horizon 2 (expected).
- Ensemble improves stability across folds.
- LightGBM performs strongest on LogLoss and MAE.
- CatBoost slightly stronger on AUC in some folds.

Production Inference

Final models trained on full dataset.

Predictions generated for:

- Purchase probability (H1 + H2)
- Quantity prediction (H1 + H2)

Reproducibility

- Fixed random seeds
- Deterministic feature pipeline
- Consistent categorical encoding
- GPU CatBoost with fixed device

Conclusion

This solution achieves strong performance through:

- Deep temporal feature engineering
- Strict leakage control
- Realistic validation simulation
- Multi-model ensembling
- Unified multi-horizon learning